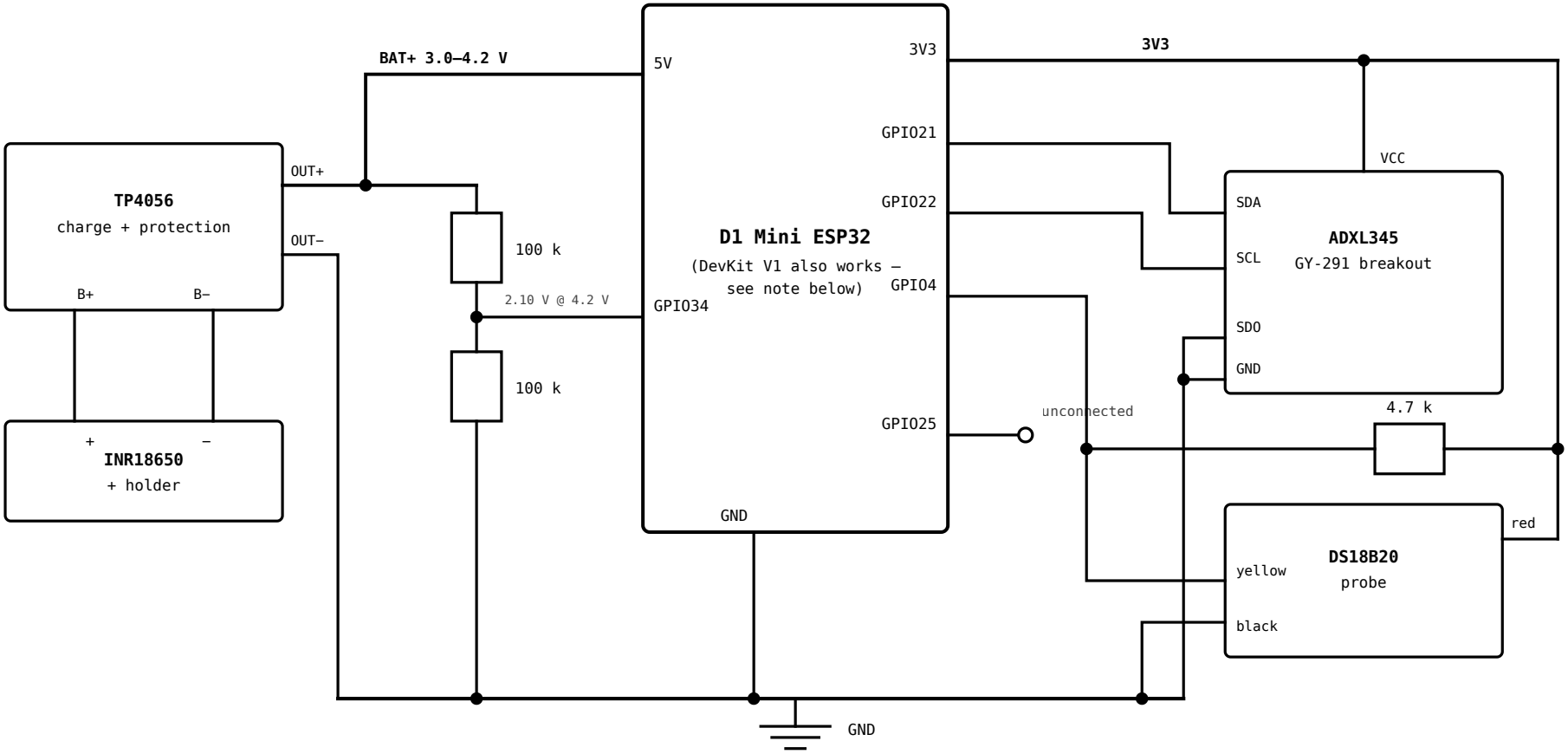
Prototype wiring — D1 Mini ESP32 + ADXL345 + INR18650

The bench rig, drawn at module level. This is *not* the custom PCB schematic (`schematic-gravity-module.html`) — that one shows bare SMD parts you can't breadboard. Every line below is a wire you physically run.



Every line is a wire. A filled dot is a junction; lines crossing without a dot are not connected (the 4.7 kΩ pull-up crosses the ADXL345's ground drop at one point). **SDO is tied to GND** — that is what puts the ADXL345 at address `0x53`, which is the address the firmware expects.

GPIO numbers, not silkscreen labels. Board revisions and clones print these pins differently (`IO4`, `D4`, plain `4`). The firmware addresses GPIO numbers; match those. Both the D1 Mini ESP32 and the DevKit V1 break out all five.

On a DevKit V1 this power chain does not work. Its AMS1117 regulator needs >4.4 V in to hold 3.3 V out, and a cell is 4.2 V falling — you get brownouts during WiFi transmit that look like random resets. Either regulate the cell to 3.3 V with an HT7333/MCP1700 and feed `3V3` directly, or use the D1 Mini, whose regulator runs from a cell as drawn. **Never feed the raw cell into `3V3`** (4.2 V against a 3.6 V maximum), and never have USB connected at the same time as an external 3V3 feed.

Connections

From	To	Why
`18650 + / −`	TP4056 `B+` / `B−`	Cell into the charger
TP4056 `OUT+`	Board `5V`	Through the board's own regulator — not `3V3`
TP4056 `OUT+`	100 k → GPIO34 → 100 k → GND	Battery sense. 2.10 V at a 4.2 V cell, matching the firmware's ×2
TP4056 `OUT−`	GND bus	Common ground
`GPIO21 / GPIO22`	ADXL345 `SDA` / `SCL`	I²C
ADXL345 `VCC`	Board `3V3`	
ADXL345 `SDO`	GND	**Sets address 0x53.** Floating or high gives 0x1D and the sketch reports no samples
`GPIO4`	DS18B20 yellow	1-Wire data
`GPIO4`	4.7 kΩ → `3V3`	Bus pull-up. Some probe modules already have one — don't fit two
`DS18B20 red / black`	`3V3` / GND	
`GPIO25`	— nothing	Divider gating isn't fitted on a breadboard; the firmware toggles it harmlessly

None of this is needed for the axis test. Board + ADXL345 + USB cable is enough to confirm the sketch prints ~0° upright and ~90° flat — the assumption the whole firmware rests on. Wire the battery and DS18B20 afterwards.